SLA, SLO, and SLI - Simple Detailed Guide

This document explains SLA, SLO, and SLI in simple language with examples.

SLA (Service Level Agreement)

- SLA is a formal promise or contract between a service provider and a customer.
- It defines what level of service the customer will receive.
- It may include penalties, refunds, or service credits if targets are missed.
- Example: 99.9% uptime every month.

SLO (Service Level Objective)

- SLO is an internal target set by the company for service performance.
- It helps teams know what they should achieve.
- SLOs are usually stricter than SLAs.
- Example: Keep uptime at 99.95% so the SLA of 99.9% is safe.

SLI (Service Level Indicator)

- SLI is the actual measurement of performance.
- It is the metric used to check reliability.
- Examples include uptime, latency, error rate, success rate.

Relationship

- SLI = What we measure.
- SLO = Goal for the measurement.
- SLA = Promise to customer.

Real Example - Online Shopping Website

- SLI: Checkout success rate is 99.7%.
- SLO: Maintain checkout success rate above 99.5%.
- SLA: If it drops below 99%, customers receive credits.

Common Metrics

- Availability/Uptime

- Response Time (Latency)
- Error Rate
- Throughput
- Support Response Time

Why Important

- Improves customer trust.
- Helps teams focus on reliability.
- Creates measurable goals.
- Supports better monitoring and alerts.

Interview Answer

- SLA is the customer agreement, SLO is the internal target, and SLI is the measured metric used to track performance.